



APS MINDS

ARCTES — Autonomous Research & Technology Intelligence

Intelligence That Thinks. Research That Moves.

Project Report

Team: Archi Sharma · Parth Goyal · Sonu Sharma

Institution: Quantum University, Roorkee

Domain: Artificial Intelligence · Generative AI ·
Autonomous Systems · Research Intelligence



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1. Executive Summary

APS Minds is an AI-powered research and technology intelligence platform built around **ARCTES — Autonomous Research & Technology Intelligence**.

The platform explores a shift from traditional reactive AI assistants toward a more proactive and autonomous intelligence model.

Most conventional AI systems follow:

USER → PROMPT → AI → RESPONSE

APS Minds is designed around a broader intelligence cycle:

DISCOVER → DECIDE → REMEMBER → CREATE → PUBLISH
 ↑ ↓
 └── CONTINUE ─┘

The current implementation provides the foundation of this vision through a modern web platform, ARCTES conversational AI, authentication, dashboard infrastructure, AI-powered interactions, and the conceptual foundation for autonomous intelligence.

The long-term architecture introduces specialized AI responsibilities for:

- Information discovery
- Relevance evaluation
- Knowledge retention
- Content generation
- Publishing

These responsibilities are represented by the planned **Scout, Editor, Archivist, Author and Publisher** agent architecture.

Importantly, these specialized agents represent the **planned multi-agent architecture and future autonomous layer**; they should not be interpreted as fully autonomous production agents in the current implementation.

The project therefore focuses on building the foundation first and creating a scalable path toward autonomous research intelligence.

2. Problem Statement

The Information Overload Problem

Technology is evolving at an unprecedented pace.

Every day, new:

- AI models
- Research papers
- Cybersecurity developments
- Open-source projects
- Developer tools
- Frameworks
- Scientific discoveries
- Industry announcements

are introduced.

The challenge is no longer simply **finding information**.

The real challenge is:

Finding what matters, understanding why it matters, remembering it, and turning it into useful intelligence.

Traditional Research Workflow

A user typically has to:

Search



Read multiple sources



Compare information



Determine relevance



Understand the topic



Remember previous findings



Create an output

This process is:

- Time-consuming

- Repetitive
 - User-dependent
 - Difficult to maintain continuously
-

Limitation of Conventional AI Assistants

Traditional AI assistants have significantly improved access to information, but their interaction model is still largely reactive:

User asks

↓

AI responds

↓

Interaction ends

The AI generally waits for another instruction.

APS Minds explores a different question:

What if an AI could continuously participate in the research process instead of waiting for every individual instruction?

3. Our Solution

APS Minds introduces **ARCTES**, a technology-focused AI intelligence system designed to evolve toward autonomous research.

The platform is structured around five conceptual stages:

01 — DISCOVER

Identify potentially valuable information and emerging technology signals.

02 — DECIDE

Evaluate relevance, importance, originality and potential value.

03 — REMEMBER

Maintain knowledge of previous research and decisions.

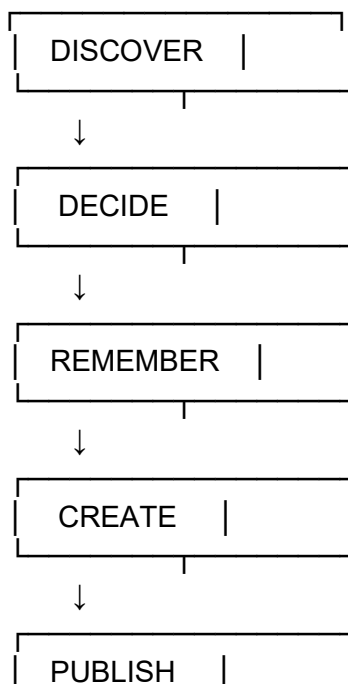
04 — CREATE

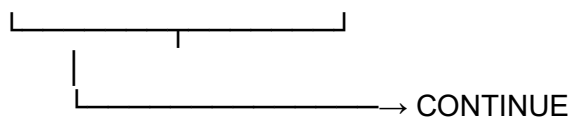
Transform validated intelligence into useful research or content.

05 — PUBLISH

Prepare qualified outputs for users or future publishing workflows.

The result is a continuous intelligence model:





4. What is ARCTES?

ARCTES

Autonomous Research & Technology Intelligence

ARCTES is the core AI intelligence layer of APS Minds.

It is designed as a technology-focused AI system rather than a generic chatbot.

The current ARCTES interface allows users to interact with AI for research-oriented tasks such as:

- Exploring AI trends
 - Understanding technology
 - Learning cybersecurity concepts
 - Generating technology-focused content
 - Developing content strategies
 - Asking custom technical questions
-

ARCTES Personality

ARCTES is designed to be:

- Analytical
- Technical
- Research-oriented
- Professional
- Concise but explanatory
- Technology-focused
- Context-aware
- Autonomous in its long-term design

Its central philosophy is:

Don't just answer. Develop intelligence.

5. Project Objectives

APS Minds has been designed around the following objectives.

Objective 1 — Build an AI Research Interface

Provide users with a dedicated AI workspace for interacting with ARCTES.

Objective 2 — Simplify Technology Research

Reduce the effort required to understand complex and rapidly changing technical information.

Objective 3 — Introduce Autonomous AI Concepts

Move beyond a simple chatbot architecture toward a continuous intelligence system.

Objective 4 — Build a Multi-Agent Foundation

Separate future autonomous responsibilities into specialized agents.

Objective 5 — Introduce Persistent Intelligence

Create an architecture where previous knowledge can influence future decisions.

Objective 6 — Create a Scalable Platform

Build the system in a modular way so additional AI capabilities can be introduced without redesigning the entire application.

6. Current Platform

The current APS Minds platform provides the foundation for the ARCTES ecosystem.

Main Components

Landing Page

Introduces:

- APS Minds
- ARCTES
- Autonomous intelligence concept
- Platform capabilities
- Project vision

ARCTES

Dedicated AI interaction environment where users can communicate with the system.

Authentication

Provides:

- User registration
- Login
- Access control
- User-specific application access

Dashboard

Provides a centralized environment for accessing platform functionality.

ARCTES Signals

Introduces the concept of technology intelligence and emerging signals.

Autonomous Engine Concept

Provides the conceptual foundation for future autonomous research workflows.

7. ARCTES User Experience

The ARCTES interface is designed to make complex AI capabilities accessible without requiring users to understand the underlying architecture.

Users can interact through natural language.

The interface also provides guided prompts for common use cases.

Example Prompts

Research the latest AI trends.

Create a technology article.

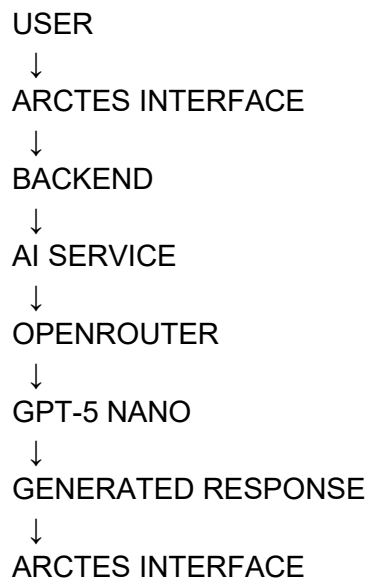
Explain a cybersecurity concept.

Generate a technology content strategy.

Users are not restricted to predefined prompts.

They can also enter custom questions and research requests.

Interaction Flow



This architecture keeps the user experience simple while allowing the backend to manage the AI integration.

8. Autonomous Intelligence Concept

The defining concept of APS Minds is the transition from **reactive AI** to **continuous intelligence**.

Traditional AI

Question



Answer

APS Minds

Discover



Evaluate



Remember



Create



Publish



Discover Again

The objective is not to claim that the entire autonomous loop is already operating in production.

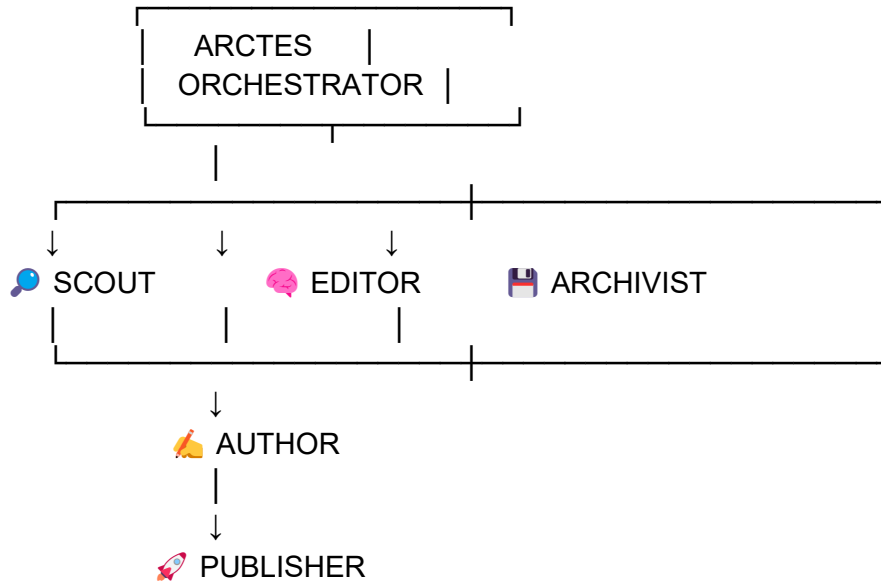
Instead, the current platform establishes the **foundation and architecture required to build this capability progressively**.

This approach allows the project to evolve from a functional AI platform into a more autonomous research system.

9. Multi-Agent Architecture

The future ARCTES architecture is based on specialized AI responsibilities.

Instead of placing every responsibility inside one large agent, the system can divide the intelligence pipeline into specialized components.



The orchestrator represents the coordination layer.

The specialized agents represent the planned autonomous responsibilities.

10. Planned Specialized Agents

Scout Agent

Purpose

Discover potentially valuable information.

Potential Responsibilities

- Monitor emerging AI developments
- Identify technology trends
- Discover research opportunities
- Track cybersecurity developments
- Surface open-source innovations
- Collect technology signals

Core Question

What is happening?

Editor Agent

Purpose

Evaluate discovered information.

Potential Responsibilities

- Determine relevance
- Evaluate originality
- Assess importance
- Prioritize information
- Detect duplicate topics
- Determine editorial value

Core Question

What actually matters?



Archivist Agent

Purpose

Maintain persistent intelligence.

Potential Responsibilities

- Store previous research
- Maintain historical context
- Track topics already covered
- Store editorial decisions
- Prevent repetitive outputs
- Provide relevant previous context

Core Question

What do we already know?



Author Agent

Purpose

Transform validated intelligence into structured content.

Potential Responsibilities

- Generate research summaries
- Create technical articles
- Produce trend reports
- Explain technical concepts
- Generate content briefs

Core Question

How can this intelligence become useful knowledge?



Publisher Agent

Purpose

Prepare qualified outputs for delivery or future publishing.

Potential Responsibilities

- Format final content
- Apply publishing rules
- Maintain consistency
- Prepare platform-specific outputs
- Support future automated publishing

Core Question

How should this intelligence reach the user?

Important Implementation Note

The **Scout, Editor, Archivist, Author and Publisher** components represent the planned multi-agent architecture and future autonomous layer.

They are included in the system design to establish a scalable path toward autonomous operation rather than being presented as fully implemented production agents in the current version.

11. Technology Stack

Frontend

Technology	Purpose
React	Component-based UI development
Vite	Frontend development and build tooling
TypeScript	Type-safe development
Tailwind CSS	Responsive interface styling
Framer Motion	Animations and interactive transitions

Backend

Technology	Purpose
Python	Backend and AI integration
FastAPI	REST API development
SQLAlchemy	Database ORM
Alembic	Database migrations
JWT	Authentication
APScheduler	Scheduled task foundation

AI Layer

Technology	Purpose
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OpenRouter API	AI model access
OpenAI GPT-5 Nano	ARCTES response generation

Development & Collaboration

Tool	Purpose
Visual Studio Code	Development environment
Git	Version control
GitHub	Repository and collaboration
Node.js / npm	Frontend ecosystem
Python	Backend ecosystem

12. AI Integration

OpenRouter API

ARCTES currently uses **OpenRouter API** as its AI provider.

Configuration

Provider:

OpenRouter

Purpose:

ARCTES AI responses

Base URL:

<https://openrouter.ai/api/v1>

Environment Variable:

OPENROUTER_API_KEY

Current Model:

openai/gpt-5-nano

The project previously had:

openai/gpt-5

configured, but the current implementation uses:

openai/gpt-5-nano

Why OpenRouter?

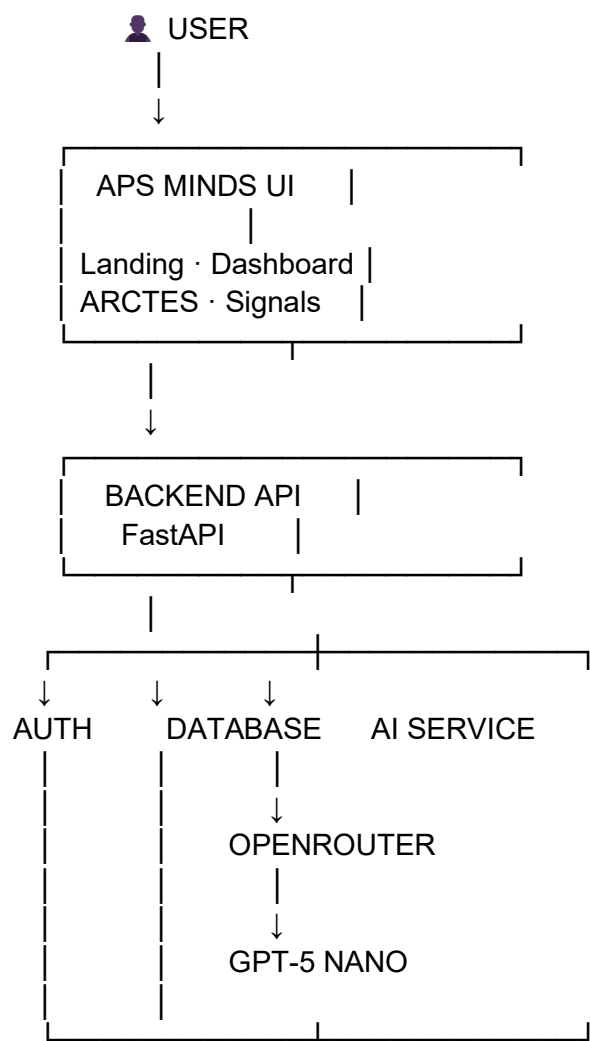
OpenRouter provides a unified interface for accessing modern AI models.

This gives APS Minds a modular AI layer where the underlying model can be changed or expanded without redesigning the entire application architecture.

This is particularly useful for an evolving multi-agent system where different tasks may eventually benefit from different models.

13. System Architecture

The current platform follows a layered architecture.



14. Authentication & Security

Security is treated as a core architectural requirement.

Authentication

The platform includes user authentication capabilities such as:

- Registration
- Login
- Protected access
- User-specific sessions

JWT-based authentication provides the foundation for securing protected API resources.

API Key Protection

The OpenRouter API key is stored through environment configuration.

OPENROUTER_API_KEY=YOUR_OPENROUTER_API_KEY

The real API key should never be:

- Hard-coded into source code
 - Exposed in frontend JavaScript
 - Committed to GitHub
 - Included in screenshots or public documentation
-

Backend AI Communication

AI provider communication is handled through the backend architecture.

This helps prevent sensitive provider credentials from being exposed to the client.

15. ARCTES Signals

ARCTES Signals represents the intelligence-discovery layer of APS Minds.

The concept is based on continuously monitoring developments that may be relevant to the technology ecosystem.

Potential signal categories include:

- Artificial intelligence
- Machine learning
- Cybersecurity
- Developer technologies
- Open-source projects
- Research breakthroughs
- Emerging technologies

The long-term objective is to transform this component into an active discovery mechanism that can continuously provide new intelligence to the autonomous pipeline.

16. Memory & Persistent Intelligence

Autonomy requires memory.

A system that discovers information but immediately forgets it cannot meaningfully build intelligence over time.

Without Memory

Discover
↓
Forget
↓
Discover Again

With Persistent Intelligence

Discover
↓
Remember
↓
Compare
↓
Evaluate
↓
Build Upon Previous Knowledge

The planned Archivist layer is therefore important to the long-term ARCTES architecture.

It can eventually maintain:

- Research history
- Previous outputs
- Topics already covered
- Editorial decisions
- Rejected topics
- Monitored technologies
- Relevant context

This creates the foundation for **context-aware autonomous behavior**.

17. Current Implementation Status

The following table clearly separates the current implementation from future architecture.

Capability	Status
APS Minds Landing Page	✔ Implemented
Navigation	✔ Implemented
Features Section	✔ Implemented
About / Team Sections	✔ Implemented
Authentication Interface	✔ Implemented
User Registration	✔ Implemented
Login	✔ Implemented
Dashboard	✔ Implemented
ARCTES Chat Interface	✔ Implemented
AI Responses	✔ Implemented
OpenRouter Integration	✔ Implemented
GPT-5 Nano Integration	✔ Implemented
Suggested AI Prompts	✔ Implemented
ARCTES Signals Concept	✔ Implemented
Autonomous Engine Concept	✔ Implemented

Backend API Architecture	✓ Implemented
Database Foundation	✓ Implemented
Scout Agent	🔄 Planned
Editor Agent	🔄 Planned
Archivist Agent	🔄 Planned
Author Agent	🔄 Planned
Publisher Agent	🔄 Planned
Fully Autonomous Research Loop	🔄 Future
Automated Publishing	🔄 Future

This distinction reflects the current state of the project accurately while showing a clear path for future development.

18. Innovation

APS Minds focuses on several areas of innovation.

18.1 From Chatbot to Intelligence System

The project moves beyond the idea of AI as only a question-answering interface.

The long-term objective is:

ASK → ANSWER

becoming:

DISCOVER → UNDERSTAND → REMEMBER → ACT

18.2 Multi-Agent Architecture

Complex responsibilities are separated into specialized roles.

This allows the system to become more modular and scalable.

18.3 Persistent Intelligence

The architecture recognizes memory as a core requirement for meaningful autonomy.

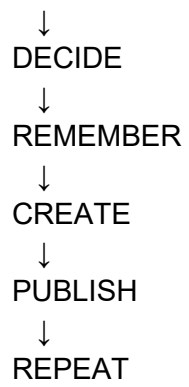
18.4 Editorial Decision Layer

The system is designed not simply to collect information, but eventually to determine whether information is relevant and worth transforming into an output.

18.5 Continuous Intelligence Loop

The project introduces a continuous operating model:

DISCOVER



This represents the central architectural vision of APS Minds.

19. Challenges & Engineering Decisions

Challenge 1 — Building a Scalable AI Architecture

A single AI service can become difficult to maintain as responsibilities increase.

Decision

Design the system around modular services and a future orchestrator-based architecture.

Challenge 2 — Model Flexibility

AI models evolve quickly.

Locking the entire system to a single provider or model can make future upgrades difficult.

Decision

Use OpenRouter as an abstraction layer for model access.

Challenge 3 — Secure API Integration

AI credentials must remain protected.

Decision

Use environment variables and backend-side AI communication.

Challenge 4 — Designing for Autonomy Without Overclaiming

Autonomous systems require multiple layers including discovery, decision-making, memory and action.

Decision

Build the functional foundation first and establish the multi-agent architecture as a structured path toward future autonomy.

20. Future Roadmap

Phase 1 — Foundation

Current Stage

- Web platform
- Authentication
- Dashboard
- ARCTES chatbot
- OpenRouter integration
- Core backend
- Initial intelligence interface

↓

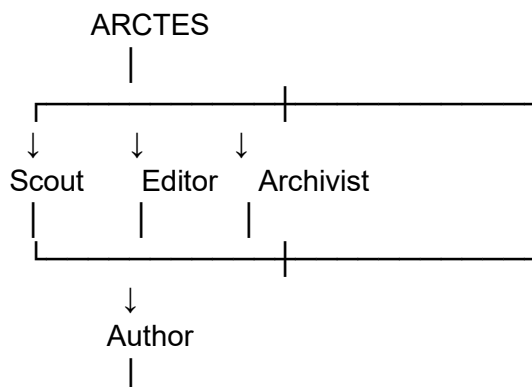
Phase 2 — Intelligence Layer

- Automated information discovery
- Research source integration
- Signal aggregation
- Persistent research memory
- Improved contextual reasoning
- Topic monitoring

↓

Phase 3 — Multi-Agent Architecture

Introduce specialized agents:



↓
Publisher

↓

Phase 4 — Autonomous Operation

Potential capabilities:

- Scheduled research
- Automated discovery
- Signal filtering
- Relevance evaluation
- Memory-based decision making
- Automated content generation
- Publishing workflows

↓

Phase 5 — Adaptive Intelligence

The system can progressively adapt based on:

- Research history
- User interests
- Previously covered topics
- Editorial decisions
- Emerging technologies
- Historical context

The ultimate objective is an intelligence system that becomes more useful through continued operation.

21. Impact & Applications

APS Minds can potentially be applied across multiple domains.



Research

Assist researchers in monitoring emerging developments.



Artificial Intelligence

Track new models, techniques, papers and AI trends.



Cybersecurity

Monitor emerging security technologies and developments.



Developer Ecosystems

Track frameworks, libraries and open-source innovations.



Technology Media

Support technology research and content generation.



Education

Help students explore complex technical topics through conversational AI.



Industry Intelligence

Assist organizations in monitoring rapidly changing technology landscapes.

22. Project Vision

APS Minds is built around a simple question:

What if an AI did not wait to be asked what to research?

ARCTES represents our answer.

Instead of treating AI as merely a conversational interface, APS Minds explores AI as an active intelligence system capable of progressively:

DISCOVERING



UNDERSTANDING



REMEMBERING



CREATING



ACTING

The ultimate vision is:

An AI that doesn't just answer questions — it continuously develops intelligence.

23. Conclusion

APS Minds establishes the foundation for an autonomous research and technology intelligence platform through ARCTES.

The current implementation combines:

- Modern web engineering
- Generative AI
- OpenRouter model integration
- Authentication
- Research-oriented interaction
- Dashboard infrastructure
- Intelligence-feed concepts
- Modular backend architecture
- Multi-agent system design

The current product demonstrates the **functional foundation** of the platform, while the multi-agent architecture provides a structured path toward future autonomous operation.

The project therefore does not attempt to claim that complete autonomy has already been achieved.

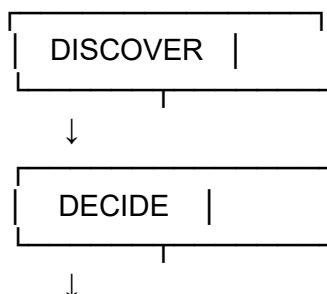
Instead, it demonstrates something equally important:

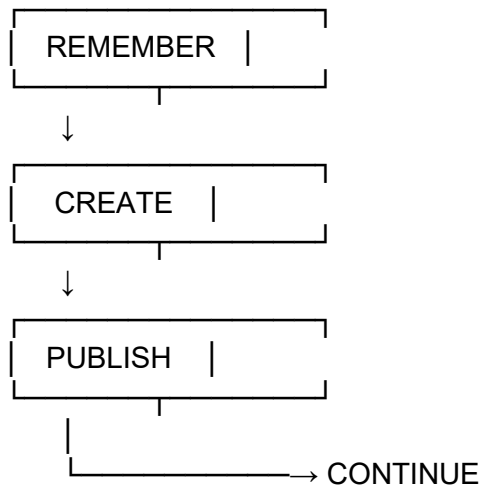
A practical foundation for building toward autonomous AI.

APS Minds moves the AI interaction model from:

USER
↓
ASK
↓
AI
↓
ANSWER

toward:





This is the foundation of **ARCTES**.

24. Technical Configuration

ARCTES AI Provider

Provider:
OpenRouter API

Purpose:
ARCTES AI Responses

Base URL:
<https://openrouter.ai/api/v1>

Environment Variable:
OPENROUTER_API_KEY

Current Model:
openai/gpt-5-nano

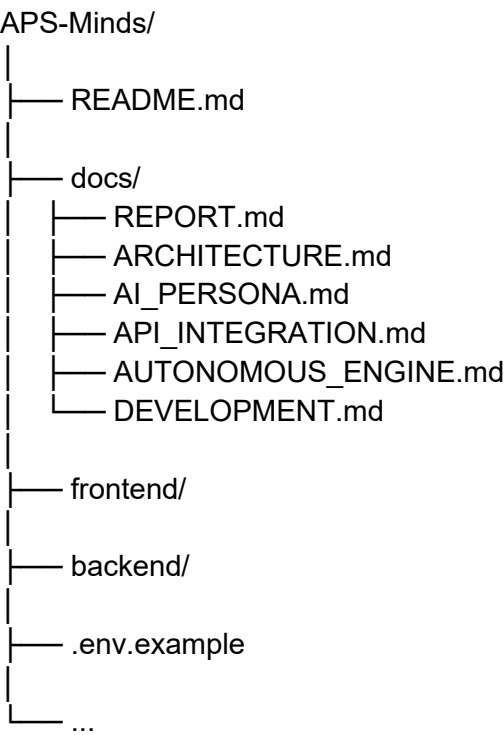
Previously Configured:
openai/gpt-5

Environment Configuration

OPENROUTER_API_KEY=YOUR_OPENROUTER_API_KEY

⚠️ Never commit the actual API key to the repository.

25. Recommended Repository Documentation



Documentation Structure

File	Purpose
README .md	Project overview and setup
REPORT .md	Complete hackathon project report
ARCHITECTURE .md	Technical architecture
AI_PERSONA .md	ARCTES identity and behavior
API_INTEGRATION .md	OpenRouter integration
AUTONOMOUS_ENGINE .md	Future autonomous architecture
DEVELOPMENT .md	Development and setup instructions



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Research That Moves.

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